TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	71.7 ± 0.94
Correlation Stability	83.7 ± 1.89
Distribution Consistency	82.7 ± 5.44
Entropy Preservation	85.3 ± 2.87
Mean Z Score Drift	0.2 ± 0.08
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	47.3 ± 7.72
Data Density	100.0 ± 0.00
Coverage Score	49.0 ± 5.89
Feature Accuracy	90.3 ± 4.11
Novelty Score	0.5 ± 0.06

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1-mini
Folds (*13)	3
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	163.122	+8.893	5.8%	OK
MDVP:Fhi	197.105	193.140	-3.965	2.0%	OK
MDVP:Flo	116.325	105.096	-11.229	9.7%	OK
MDVP:Jitter	0.006	0.005	-0.001	14.6%	Warn
MDVP:Jitter.1	0.006	0.005	-0.001	14.6%	Warn
MDVP:RAP	0.003	0.003	-0.001	18.6%	Warn
MDVP:PPQ	0.003	0.003	-0.001	15.2%	Warn
Jitter:DDP	0.010	0.008	-0.002	18.9%	Warn
MDVP:Shimmer	0.030	0.028	-0.002	6.2%	OK
MDVP:Shimmer.1	0.030	0.028	-0.002	6.2%	OK
Shimmer:APQ3	0.016	0.015	-0.001	6.2%	OK
Shimmer:APQ5	0.018	0.017	-0.001	7.4%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	71.7 ± 0.94
Correlation Stability (*2)	83.7 ± 1.89
Distribution Consistency (*3)	82.7 ± 5.44
Entropy Preservation (*4)	85.3 ± 2.87
Mean Z Score Drift (*5)	0.2 ± 0.08
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	47.3 ± 7.72

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	49.0 ± 5.89
Feature Accuracy (*10)	90.3 ± 4.11
Novelty Score (*11)	0.5 ± 0.06

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.7264957264957266	0.9658119658119658	-0.008547008547008517	
F1	0.9806435394670693	0.7998964513363719	0.9730558890222757	-0.007587650444793296	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8888888888888889	0.6239316239316239	0.8888888888888889	3.700743415417188e-17	
F1	0.9167514124293786	0.6992961530195573	0.9146916737709576	-0.0020597386584208697	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9658119658119658	0.7606837606837606	0.9316239316239315	-0.03418803418803418	
F1	0.9743589743589745	0.8155432385720246	0.9497601323407775	-0.024598842018196843	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9487179487179486	0.6495726495726496	0.94017094017094	-0.008547008547008517	
F1	0.9609674981103553	0.7193588489443222	0.9537254901960784	-0.007242007914276775	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.717948717948718	0.7521367521367521	-0.025641025641025623	
F1	0.8614267676767677	0.8084353146853146	0.8392045454545455	-0.022222222222222254	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.7521367521367521	0.8632478632478633	-0.01709401709401703	
F1	0.91232423490488	0.8145021645021645	0.9013617581511179	-0.01096247675376219	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.6666666666666666	0.8290598290598291	-0.01709401709401710	
F1	0.8918089248277927	0.7555230669265757	0.8803595363292239	-0.01144938849856880	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.5982905982905983	0.6837606837606837	-0.03418803418803418	
F1	0.749906191369606	0.675986692188497	0.7027474346242966	-0.04715875674530941	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8119658119658121	0.6923076923076924	0.8205128205128204	0.00854700854700851	
F1	0.8792936081242533	0.7937500000000001	0.8744082428292955	-0.00488536529495786	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.7777777777777778	0.8974358974358974	-0.04273504273504273	
F1	0.9570703384745131	0.8302548137899227	0.9236062861869314	-0.03346405228758168	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	178.222	202.45	141.047	0.00321	0.00321	0.00163	0.00194	0.00488	0.03759	0.03759
2	126.344	134.231	112.773	0.00448	0.00448	0.00131	0.00169	0.00393	0.02033	0.02033
3	180.978	200.125	155.495	0.00406	0.00406	0.0022	0.00244	0.00659	0.03852	0.03852
4	116.342	581.289	94.246	0.00267	0.00267	0.00115	0.00148	0.00345	0.013	0.013
5	174.688	240.005	74.287	0.0136	0.0136	0.00624	0.00564	0.01873	0.02308	0.02308
6	119.056	125.213	86.795	0.00346	0.00346	0.00169	0.0017	0.00508	0.01201	0.01201
7	138.456	162.789	110.234	0.00412	0.00412	0.00201	0.00234	0.00512	0.03145	0.03145
8	121.789	130.456	95.678	0.00356	0.00356	0.00167	0.00189	0.00401	0.02012	0.02012
9	189.234	210.567	140.123	0.00567	0.00567	0.00289	0.00312	0.00678	0.03789	0.03789
10	115.678	580.123	92.345	0.00245	0.00245	0.00112	0.00134	0.00356	0.01234	0.01234
11	175.234	238.456	75.123	0.01345	0.01345	0.00612	0.00556	0.01823	0.02289	0.02289
12	118.345	126.789	87.123	0.00345	0.00345	0.00178	0.00179	0.00501	0.01234	0.01234
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	169.774	191.759	151.451	0.01568	0.01568	0.00863	0.00946	0.02589	0.08143	0.08143
2	176.858	205.56	75.344	0.00747	0.00747	0.00418	0.0042	0.01254	0.04795	0.04795
3	113.715	116.443	96.913	0.00349	0.00349	0.00171	0.00203	0.00514	0.01472	0.01472
4	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
5	186.695	211.961	149.442	0.01872	0.01872	0.01075	0.01154	0.03225	0.0583	0.0583
6	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
7	145.234	178.456	85.123	0.00612	0.00612	0.00315	0.00345	0.00823	0.02745	0.02745
8	160.987	210.123	75.456	0.00489	0.00489	0.00234	0.00256	0.00678	0.01856	0.01856
9	130.456	145.789	110.234	0.00789	0.00789	0.00412	0.00456	0.01012	0.03456	0.03456
10	120.345	130.567	100.123	0.00567	0.00567	0.00289	0.00312	0.00789	0.02234	0.02234
11	180.123	220.456	70.789	0.00423	0.00423	0.00201	0.00234	0.00567	0.01567	0.01567
12	140.567	160.234	90.456	0.0089	0.0089	0.00456	0.00489	0.01123	0.03012	0.03012
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	150.258	154.609	75.349	0.00248	0.00248	0.00105	0.00136	0.00315	0.01131	0.01131
2	198.383	215.203	193.104	0.00212	0.00212	0.00113	0.00135	0.00339	0.01263	0.01263
3	186.695	211.961	149.442	0.01872	0.01872	0.01075	0.01154	0.03225	0.0583	0.0583
4	197.569	217.627	90.794	0.00803	0.00803	0.0049	0.00448	0.0147	0.02177	0.02177
5	193.03	208.9	80.297	0.00766	0.00766	0.0045	0.00389	0.01351	0.03044	0.03044
6	170.368	268.796	79.543	0.00571	0.00571	0.00232	0.00269	0.00696	0.03273	0.03273
7	160.123	170.456	80.234	0.00456	0.00456	0.00234	0.00267	0.00789	0.02567	0.02567
8	190.456	210.789	95.123	0.00678	0.00678	0.00345	0.00378	0.01012	0.03089	0.03089
9	140.789	160.123	75.456	0.00345	0.00345	0.00189	0.0021	0.00567	0.02012	0.02012
10	175.234	195.567	85.789	0.00567	0.00567	0.00289	0.00312	0.0089	0.02789	0.02789
11	185.678	205.123	90.234	0.00789	0.00789	0.00345	0.00389	0.01123	0.03245	0.03245
12	130.456	150.789	70.123	0.00234	0.00234	0.00123	0.00145	0.00456	0.0189	0.0189
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645